ECON 573 Problem Set 7 by Jimmy Zheng

Problem 1: ISL Chapter 6: Exercises 2 and 3

1. Exercise 2
    a. (iii) is correct. As the value of λ increases, to minimize RSS + $\lambda \sum_{i=1}^{P} |\beta_i|$, the lasso shrinks coefficients towards zero, which restricts the value that β_i can take. With more restrictions being imposed on β_i, the lasso is less flexible than the OLS. However, whether the decrease in flexibility improves prediction or not depends on the relative magnitude of the decrease in variance to the increase in bias. If the lower flexibility leads to a large decrease in variance and a small increment in bias, reducing the change of overfitting on out-of-sample data, then the prediction accuracy will be improved.
    b. (iii) is correct. As the value of λ increases, to minimize RSS + $\lambda \sum_{i=1}^{P} \beta_i^2$, the ridge regression shrinks all coefficients towards zero, which restricts the value that β_i can take. Unlike lasso, the ridge regression will include non-zero coefficients for all predictors in the model. However, there were still restrictions being imposed on β_i, so the ridge regression is less flexible than the OLS. Like the lasso method, whether the decrease in flexibility improves prediction or not depends on the relative magnitude of the decrease in variance to the increase in bias.
    c. (ii) is correct. Non-linear models are generally more flexible than the OLS, as more parameters regarding the curvature will be estimated under same number of predictors. In contrast to the analysis above, as flexibility increases, whether the prediction improves depends on the relative magnitude of the increase in variance to the decrease in bias.

2. Exercise 3
    a. (iv) the training RSS will decrease steadily. As we increase s from 0, the constrains imposed on $\sum_{i=1}^{P} |\beta_i|$ are loosening, flexibility increases, and the magnitudes of the estimated β_i's increase, approaching that of the OLS. The increase in model flexibility always lowers training RSS.
    b. (ii) is correct. When s=0, overly harsh restrictions have been β_i's, resulting in substantial bias and extremely high test RSS. Therefore, as s increases from 0, the huge reduction in bias lowers both train RSS and test RSS initially. However, when s starts getting too large, the model become overly flexible, and the increase in variance outpaces the decrease in bias, leading to overfitting, poor predictions on the test dataset, and an increase in test RSS.
    c. (iii) is correct. As we increase s from 0, the constrains imposed on $\sum_{i=1}^{P} |\beta_i|$ are loosening, flexibility increases, and more non-zero coefficients may be included. This increases the variance as the estimated parameters are less robust against changes in the test datasets.
    d. (iv) is correct. When s=0, overly harsh restrictions have been β_i's, resulting in substantial bias and extremely high test RSS. Therefore, as s increases from 0, there will be a huge reduction in bias at first. However, though the squared bias

always decreases when s become larger, the rate of bias reduction decreases eventually.

e. (v) is correct. Irreducible errors are said to be irreducible because they cannot be predicted by those covariates X's and thus will NOT be affected the model flexibility. That is also the reason why the irreducible error will always provide an upper bound on the accuracy of our prediction for Y.

Problem 2: ISL Chapter 6. Applied Exercise 8

(a) Use the rnorm() function to generate a predictor X of length n = 100, as well as a noise vector of length n = 100.

```
set.seed(1)
x = rnorm(100)
e = rnorm(100)
```

(b) Generate a response vector Y of length n = 100 according to the model

$$Y = \beta_0 + \beta_1 X + \beta_2 X^2 + \beta_3 X^3 + \epsilon$$

```
b0 = 1
b1 = 2.5
b2 = 1.5
b3 = 2

y = b0 + b1*x + b2*x^2 + b3*x^3 + e
```

(c) Use the regsubsets() function to perform best subset selection in order to choose the best model containing the predictors X, $X^{2,\ldots,X}10$. What is the best model obtained according to Cp, BIC, and adjusted R2? Show some plots to provide evidence for your answer, and report the coefficients of the best model obtained. Note you will need to use the data.frame() function to create a single data set containing both X and Y .

```
library(leaps)
data = data.frame(poly(x, 10, raw = T), y)
full = regsubsets(y ~ poly(x, 10, raw = T), data = data, nvmax = 10)

# Cp
which.min(summary(full)$cp)

## [1] 4

coef(full, 4)

##           (Intercept) poly(x, 10, raw = T)1 poly(x, 10, raw = T)2
##            1.07200775            2.88745596            1.34575641
## poly(x, 10, raw = T)3 poly(x, 10, raw = T)5
##            1.55797426            0.08072292

# BIC
which.min(summary(full)$bic)

## [1] 3
```

```r
coef(full, 3)
```

```
##           (Intercept) poly(x, 10, raw = T)1 poly(x, 10, raw = T)2
##              1.061507              2.475280              1.376209
## poly(x, 10, raw = T)3
##              2.017639
```

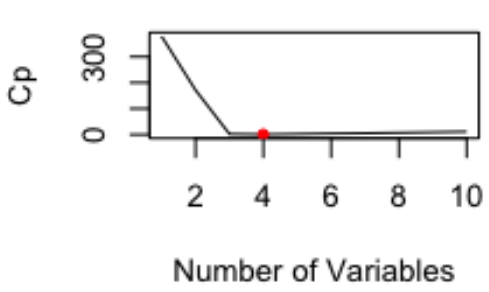
```r
# adj R^2
which.max(summary(full)$adjr2)
```

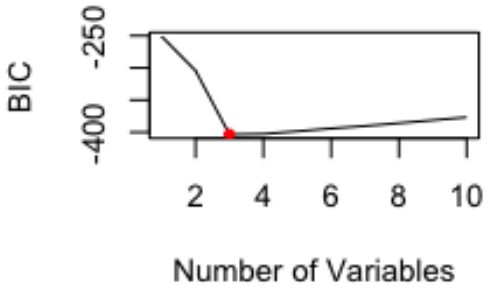
```
## [1] 4
```

```r
coef(full, 4)
```

```
##           (Intercept) poly(x, 10, raw = T)1 poly(x, 10, raw = T)2
##            1.07200775            2.88745596            1.34575641
## poly(x, 10, raw = T)3 poly(x, 10, raw = T)5
##            1.55797426            0.08072292
```

```r
# Plots
par(mfrow=c(2,2))
plot(summary(full)$cp, xlab = "Number of Variables", ylab = "Cp", type = "l")
points(4, summary(full)$cp[4], pch = 20, col = "red")
plot(summary(full)$bic, xlab = "Number of Variables", ylab = "BIC", type = "l
")
points(3, summary(full)$bic[3], pch = 20, col = "red")
plot(summary(full)$adjr2, xlab = "Number of Variables", ylab = "Adjusted R^2
", type = "l")
points(4, summary(full)$adjr2[4], pch = 20, col = "red")
```







I choose the best model selected by BIC

```
coef(full, 3)
```

```
##           (Intercept) poly(x, 10, raw = T)1 poly(x, 10, raw = T)2
##              1.061507              2.475280              1.376209
## poly(x, 10, raw = T)3
##              2.017639
```

(d) Repeat (c), using forward stepwise selection and also using backwards stepwise selection. How does your answer compare to the results in (c)?

```
fwd = regsubsets(y ~ poly(x, 10, raw = T), data = data, nvmax = 10, method = "forward")
bwd = regsubsets(y ~ poly(x, 10, raw = T), data = data, nvmax = 10, method = "backward")
```

Best Models from Forward Selections

```
# Cp
which.min(summary(fwd)$cp)
```

```
## [1] 4
```

```
coef(fwd, 4)
```

```
##           (Intercept) poly(x, 10, raw = T)1 poly(x, 10, raw = T)2
##            1.07200775            2.88745596            1.34575641
## poly(x, 10, raw = T)3 poly(x, 10, raw = T)5
##            1.55797426            0.08072292

# BIC
which.min(summary(fwd)$bic)

## [1] 3

coef(fwd, 3)

##           (Intercept) poly(x, 10, raw = T)1 poly(x, 10, raw = T)2
##              1.061507              2.475280              1.376209
## poly(x, 10, raw = T)3
##              2.017639

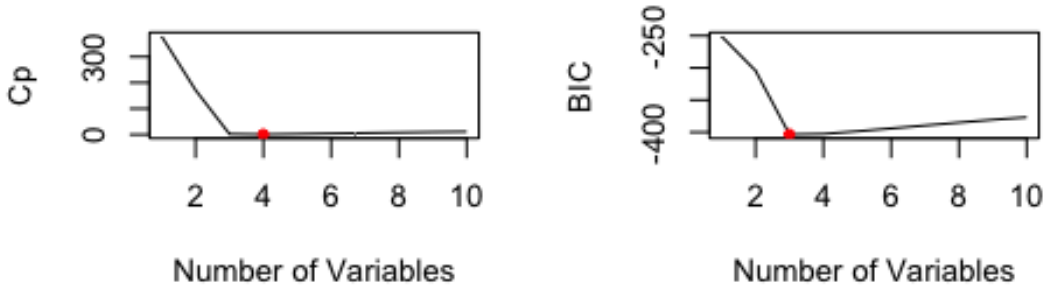
# adj R^2
which.max(summary(fwd)$adjr2)

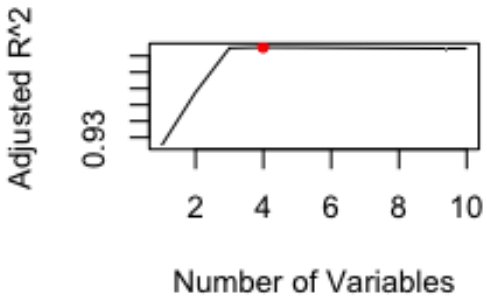
## [1] 4

coef(fwd, 4)

##           (Intercept) poly(x, 10, raw = T)1 poly(x, 10, raw = T)2
##            1.07200775            2.88745596            1.34575641
## poly(x, 10, raw = T)3 poly(x, 10, raw = T)5
##            1.55797426            0.08072292

# Plots
par(mfrow=c(2,2))
plot(summary(fwd)$cp, xlab = "Number of Variables", ylab = "Cp", type = "l")
points(4, summary(fwd)$cp[4], pch = 20, col = "red")
plot(summary(fwd)$bic, xlab = "Number of Variables", ylab = "BIC", type = "l
")
points(3, summary(fwd)$bic[3], pch = 20, col = "red")
plot(summary(fwd)$adjr2, xlab = "Number of Variables", ylab = "Adjusted R^2",
 type = "l")
points(4, summary(fwd)$adjr2[4], pch = 20, col = "red")
```





The best model selected by BIC.

```
coef(fwd, 3)
```

```
##           (Intercept) poly(x, 10, raw = T)1 poly(x, 10, raw = T)2
##              1.061507              2.475280              1.376209
## poly(x, 10, raw = T)3
##              2.017639
```

Best Models from Backward Selections

```
which.min(summary(bwd)$cp)
```

```
## [1] 7
```

```
coef(bwd, 7)
```

```
##            (Intercept)  poly(x, 10, raw = T)1  poly(x, 10, raw = T)3
##            1.225183528            2.819815184            1.746569722
##  poly(x, 10, raw = T)4  poly(x, 10, raw = T)6  poly(x, 10, raw = T)8
##            2.309845760           -1.406967677            0.335036364
##  poly(x, 10, raw = T)9 poly(x, 10, raw = T)10
##            0.002072078           -0.026883781
```

```r
# BIC
which.min(summary(bwd)$bic)
```

```
## [1] 5
```

```r
coef(bwd, 5)
```

```
##           (Intercept) poly(x, 10, raw = T)1 poly(x, 10, raw = T)3
##            1.33104498           2.59431100           1.95493979
## poly(x, 10, raw = T)4 poly(x, 10, raw = T)6 poly(x, 10, raw = T)8
##            1.14483293          -0.33839793           0.03269351
```

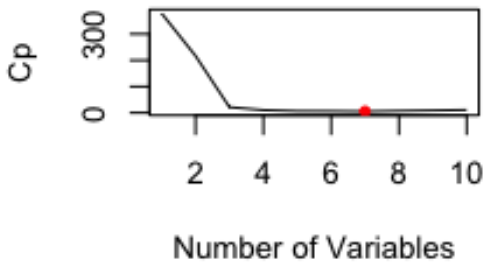
```r
# adj R^2
which.max(summary(bwd)$adjr2)
```

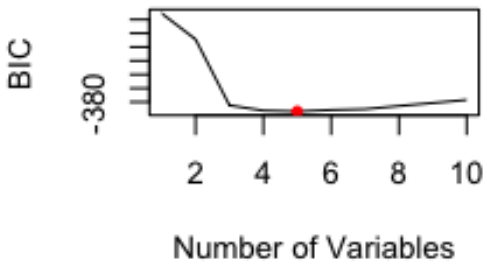
```
## [1] 7
```

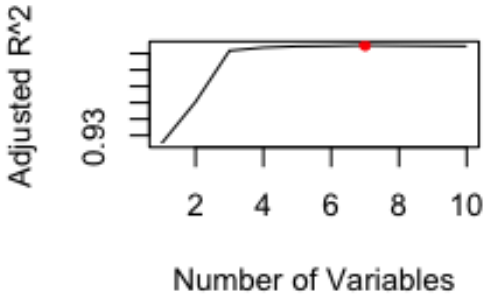
```r
coef(bwd, 7)
```

```
##           (Intercept)  poly(x, 10, raw = T)1  poly(x, 10, raw = T)3
##           1.225183528            2.819815184            1.746569722
##  poly(x, 10, raw = T)4  poly(x, 10, raw = T)6  poly(x, 10, raw = T)8
##           2.309845760           -1.406967677            0.335036364
##  poly(x, 10, raw = T)9 poly(x, 10, raw = T)10
##           0.002072078           -0.026883781
```

```r
# Plots
par(mfrow=c(2,2))
plot(summary(bwd)$cp, xlab = "Number of Variables", ylab = "Cp", type = "l")
points(7, summary(bwd)$cp[7], pch = 20, col = "red")
plot(summary(bwd)$bic, xlab = "Number of Variables", ylab = "BIC", type = "l
")
points(5, summary(bwd)$bic[5], pch = 20, col = "red")
plot(summary(bwd)$adjr2, xlab = "Number of Variables", ylab = "Adjusted R^2",
 type = "l")
points(7, summary(bwd)$adjr2[7], pch = 20, col = "red")
```







The best model selected by BIC

```
coef(bwd, 5)
```

```
##           (Intercept) poly(x, 10, raw = T)1 poly(x, 10, raw = T)3
##            1.33104498           2.59431100           1.95493979
## poly(x, 10, raw = T)4 poly(x, 10, raw = T)6 poly(x, 10, raw = T)8
##            1.14483293          -0.33839793           0.03269351
```

For this simulated series, we luckily arrive at the same best models by using both regsubset and forward selections based on Cp, BIC and Adjusted R^2. However, the models selected by backward selection are different from both of them. In this example, not only does backward selection include more varianbles, but also higher orders of X's, such as X^10, X^9, and X^8.

(e) Now fit a lasso model to the simulated data, again using X, $X^{2,\ldots,X}10$ as predictors. Use cross-validation to select the optimal value of . Create plots of the cross-validation error as a function of . Report the resulting coefficient estimates, and discuss the results obtained.

```
library(glmnet)
```

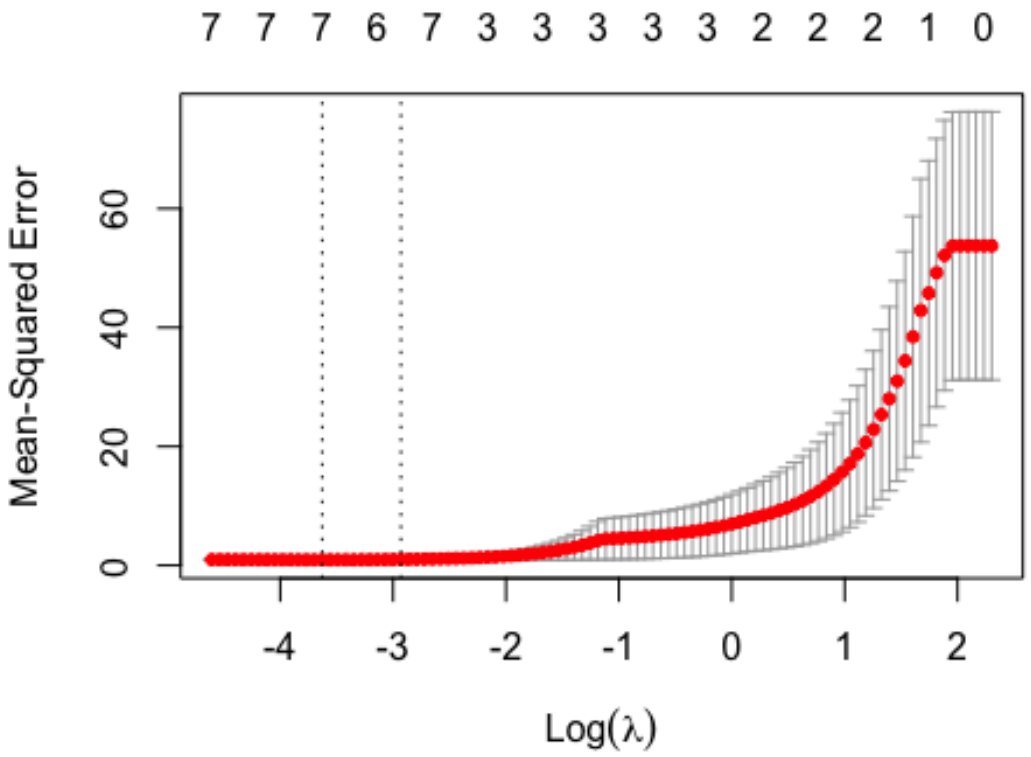
```
## Loading required package: Matrix
```

```
## Loaded glmnet 4.1-4
```

```
x_mat = model.matrix(y~poly(x, 10, raw = T),data=data)[,-1]
y = data$y
grid = 10^seq(1, -2, length = 100)
colnames(x_mat) = c("X", "X^2", "X^3", "X^4", "X^5", "X^6", "X^7", "X^8", "X^
9", "X^10")

set.seed(1)
train = sample(1:nrow(x_mat), nrow(x_mat)/2)
test = (-train)
y.test = y[test]

mod.lasso1 = glmnet(x_mat[train,], y[train], alpha = 1,  lambda = grid)
cv.out = cv.glmnet(x_mat[train,], y[train], alpha = 1, lambda = grid)
plot(cv.out)
```




```
# best value of lambda
bestlam = cv.out$lambda.min

# report the coefficients
out = glmnet(x_mat, y, alpha = 1, lambda = grid)
lasso1.coef = predict(out, type = "coefficients", s = bestlam)[1:11,]
lasso1.coef
```

```
## (Intercept)           X         X^2         X^3         X^4         X^5
## 1.143738781 2.718361295 1.172012523 1.792007429 0.038003345 0.004487970
##         X^6         X^7         X^8         X^9        X^10
## 0.000000000 0.006263942 0.000000000 0.000000000 0.000000000

# non-zero coefficients
lasso1.coef[lasso1.coef!=0]

## (Intercept)           X         X^2         X^3         X^4         X^5
## 1.143738781 2.718361295 1.172012523 1.792007429 0.038003345 0.004487970
##         X^7
## 0.006263942
```

The lasso method produce a relatively sparse model with only 6 out of 10 variables having non-zero estimated coefficients, which is close to the true model with order 3. The intercept and coefficients of X, X^2, and X^3 are similar to the true parameters b0, b1, .., b3. Although it produces non-zero coefficients for X^5 and X^7, their coefficients are approximately zero, which is consistent with the model simulated.

(f) Now generate a response vector Y according to the model

$$Y = \beta_0 + \beta_7 X^7 + \epsilon$$

and perform best subset selection and the lasso. Discuss the results obtained.

```
b0_new = 1
b7_new = 2
y_new = b0_new + b7_new*x^7 + e

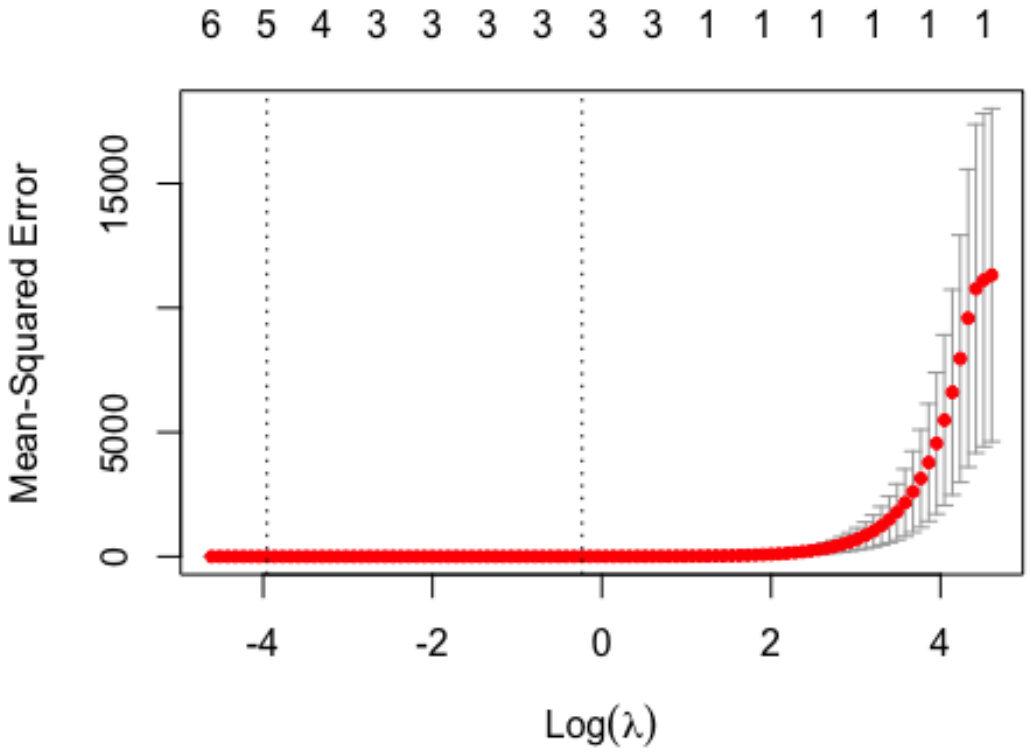
# Best Subset Selection
data_new = data.frame(poly(x, 10, raw = T), y_new)
full_new = regsubsets(y_new ~ ., data = data_new, nvmax = 10)
coef(full_new, which.min(summary(full_new)$bic))

## (Intercept)          X7
##   0.9589402   2.0007705

# Lasso
x_mat_new = model.matrix(y_new~poly(x, 10, raw = T),data=data_new)[,-1]
y = data_new$y_new
grid = 10^seq(2, -2, length = 100)
colnames(x_mat_new) = c("X", "X^2", "X^3", "X^4", "X^5", "X^6", "X^7", "X^8",
 "X^9", "X^10")

set.seed(1)
train = sample(1:nrow(x_mat_new), nrow(x_mat_new)/2)

mod.lasso2 = glmnet(x_mat_new[train,], y[train], alpha = 1,  lambda = grid)
cv.out2 = cv.glmnet(x_mat_new[train,], y[train], alpha = 1, lambda = grid)
plot(cv.out2)
```




```
bestlam2 = cv.out2$lambda.min

out2 = glmnet(x_mat_new, y, alpha = 1, lambda = grid)
lasso1.coef2 = predict(out2, type = "coefficients", s = bestlam2)[1:11,]
lasso1.coef2[lasso1.coef2!=0]

##   (Intercept)           X^2           X^7           X^8           X^9
##  1.073945e+00 -1.587727e-01  1.993690e+00  6.951978e-05  1.365103e-03
##          X^10
##  6.257630e-05
```

For this simulated model, the best model selected by regsubset based on BIC surprisingly outperforms the one estimated by the Lasso method. The model from regsubset includes X^7 only and has almost identical estimates of beta_0 and beta_7. However, the lasso method also performs pretty well. It produces nearly negligible coefficients for variables other than X^7, and the estimated coefficients for beta _0 and beta_7 are similar to the true parameters.